Motivación al método de regresión

Econometría - FCE Austral

Ricardo Pasquini

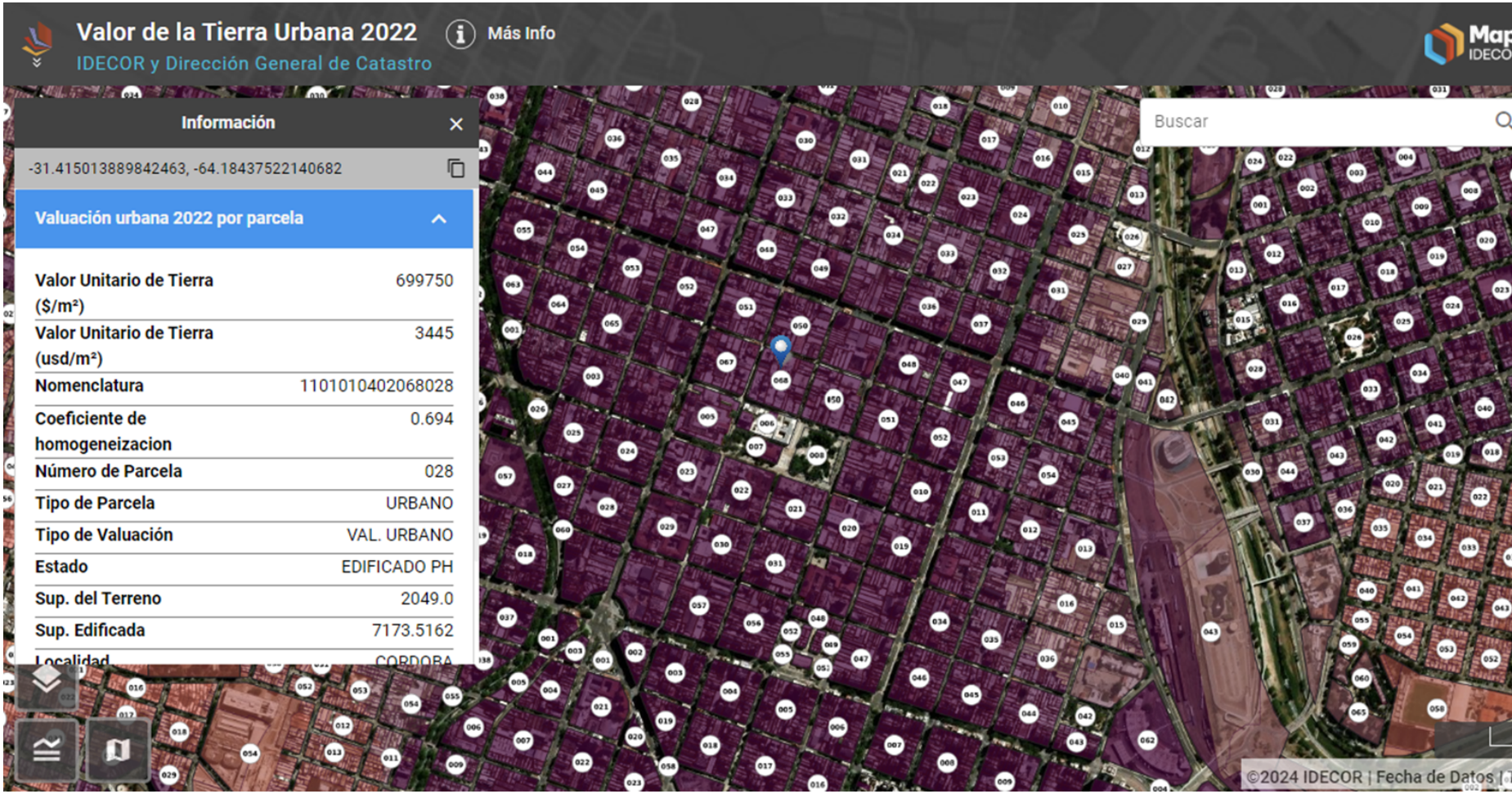
March 2, 2026

Usos de los Modelos de Regresión

1. Predicción
2. Extrapolación (una forma de predicción)
3. Inferencia causal
   - Experimental
   - Cuasi-experimental y observacional

Predicción

Valuación de terrenos



https://mapascordoba.gob.ar/viewer/mapa/401

Predicción

Recomendación de valor de alquiler

Startups

Airbnb Adds A Pricing Recommendation Tool For Renters

Matthew Lynley / 10:34 AM PDT • June 4, 2015

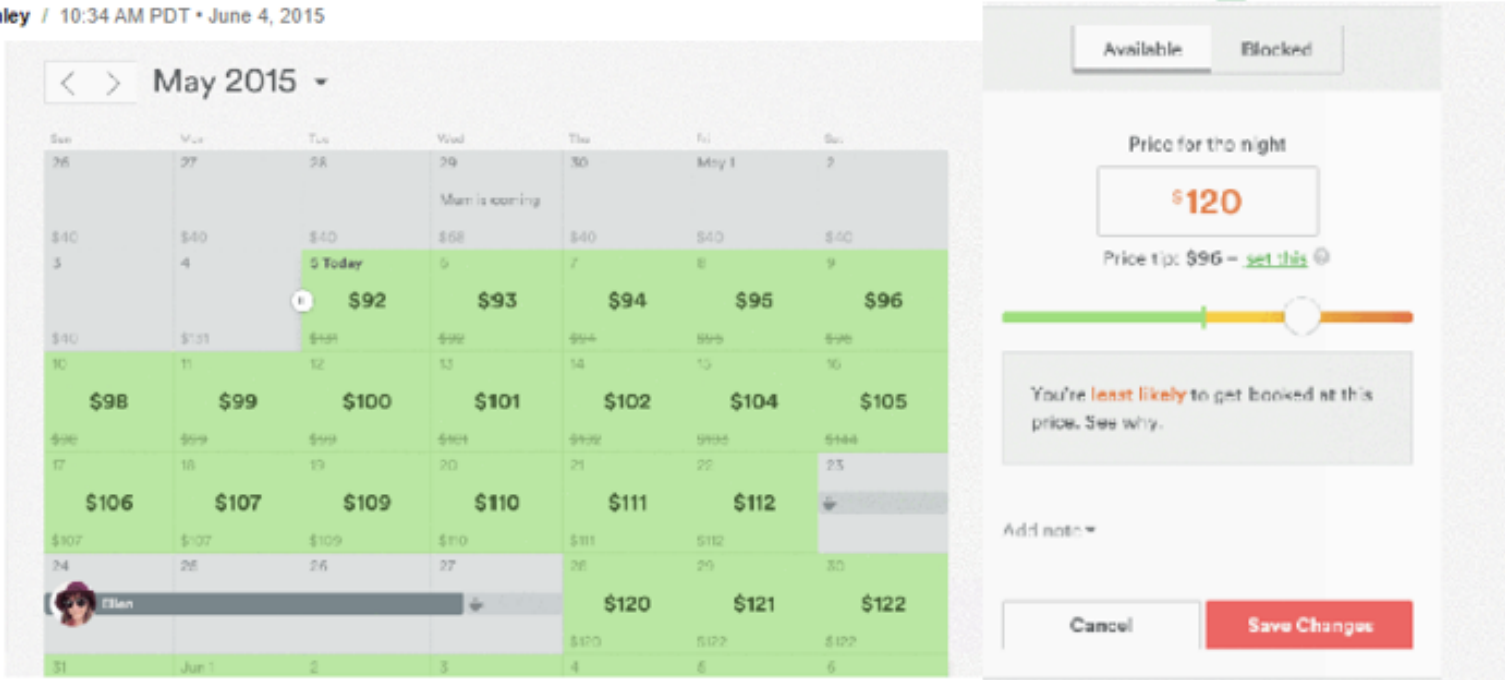


Image Credits: Airbnb

https://techcrunch.com/2015/06/04/airbnb-adds-a-pricing-recommendation-tool-for-renters/

Extrapolación

Extrapolación entre subgrupos

Forecasting elections with non-representative polls

Wei Wang[a,*], David Rothschild[b], Sharad Goel[b], Andrew Gelman[a,c]

A B S T R A C T

Election forecasts have traditionally been based on representative polls, in which randomly sampled individuals are asked who they intend to vote for. While representative polling has historically proven to be quite effective, it comes at considerable costs of time and money. Moreover, as response rates have declined over the past several decades, the statistical benefits of representative sampling have diminished. In this paper, we show that, with proper statistical adjustment, non-representative polls can be used to generate accurate election forecasts, and that this can often be achieved faster and at a lesser expense than traditional survey methods. We demonstrate this approach by creating forecasts from a novel and highly non-representative survey dataset: a series of daily voter intention polls for the 2012 presidential election conducted on the Xbox gaming platform. After adjusting the Xbox responses via multilevel regression and poststratification, we obtain estimates which are in line with the forecasts from leading poll analysts, which were based on aggregating hundreds of traditional polls conducted during the election cycle. We conclude by arguing that non-representative polling shows promise not only for election forecasting, but also for measuring public opinion on a broad range of social, economic and cultural issues.

Extrapolación

Extrapolación entre subgrupos

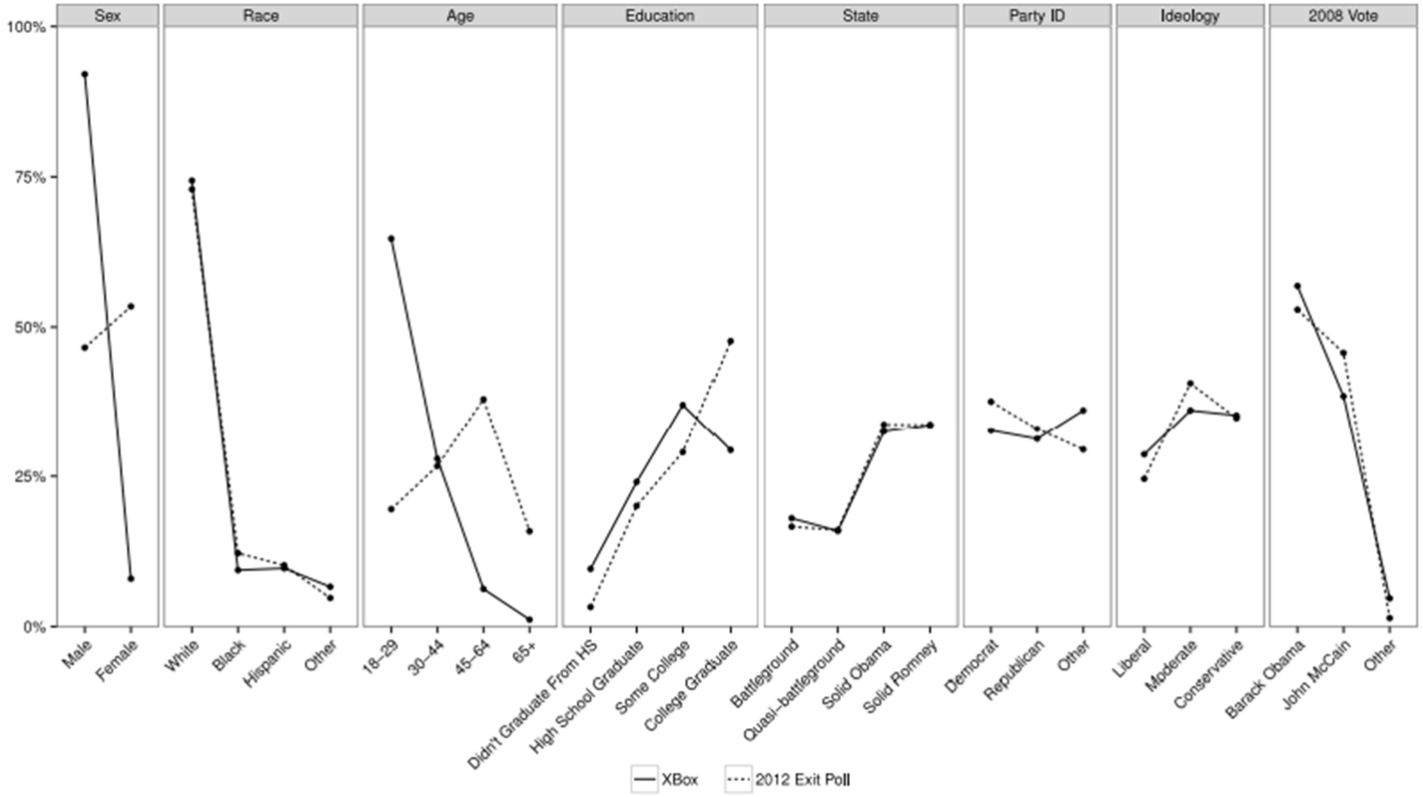
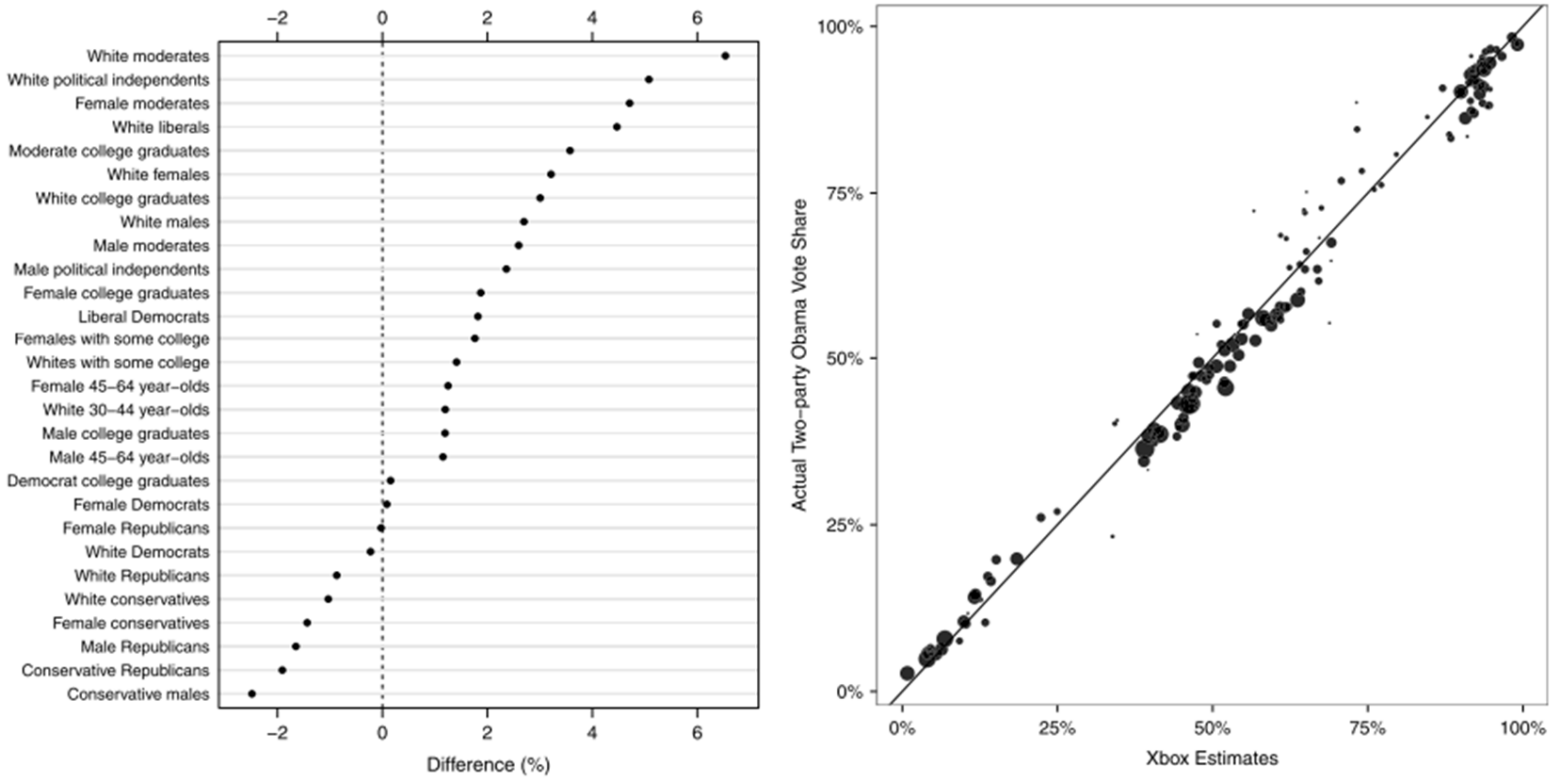


Fig. 1. A comparison of the demographic, partisan, and 2008 vote distributions in the Xbox dataset and the 2012 electorate (as measured by adjusted exit polls). As one might expect, the sex and age distributions exhibit considerable differences.

Extrapolación

Extrapolación entre subgrupos

Fig. 5. Comparison of the two-party Obama vote share for various demographic subgroups, as estimated from the 2012 national exit poll and from the Xbox data on the day before the election.



Inferencia Causal

Análisis de experimentos

ELSEVIER

Journal of Urban Economics

Volume 142, July 2024, 103612

JUE insight: The unintended effect of Argentina's subsidized homeownership lottery program on intimate partner violence

Bruno Cardinale Lagomarsino [a], Martin A. Rossi [b]

Abstract

We study a natural experiment in Argentina, where low-income women were selected through a lottery system to receive a house and a heavily subsidized long-term mort- gage. We exploit the random assignment to estimate the causal link between sub- sidized homeownership programs and intimate partner violence (IPV). Our analysis utilizes administrative records of the population of women applicants to assess the im- pact of homeownership on IPV, differentiating between women under joint-ownership contracts with their partners and those under single-ownership contracts. We find that the program causes an increase in IPV for women under joint-ownership contracts and a decrease in IPV for women under single-ownership contracts. Our results highlight the importance of considering the design of subsidized homeownership programs and explicitly incorporating measures to facilitate exit from conflicting relationships.

https://www.sciencedirect.com/science/article/abs/pii/S0094119023000827

Inferencia Causal

Análisis de experimentos

	Dependent variable: IPV							
	Joint ownership		Women ownership		Joint ownership		Women ownership	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Beneficiary	0.173***	0.196***	-0.113	-0.116*	0.072	0.069	-0.608***	-0.501***
	(0.049)	(0.047)	(0.070)	(0.069)	(0.122)	(0.127)	(0.172)	(0.146)
Control group mean	0.114	0.114	0.239	0.239	0.386	0.386	0.887	0.887
Control variables	No	Yes	No	Yes	No	Yes	No	Yes
Sample	Admin	Admin	Admin	Admin	Survey	Survey	Survey	Survey
Observations	290	290	147	147	85	85	37	37

Notes: Heteroskedastic-robust standard errors in parentheses. The 437 observations correspond to the sample of eligible non-attritor women. Dependent variable equals 1 if the women reported IPV in the administrative data and zero otherwise in columns 1 to 4. Dependent variable is equal to 1 if the woman reported IPV in the survey and zero otherwise in columns 5 to 8. The main independent variable equals 1 if the woman is a beneficiary of the program and zero otherwise. The instrument equals 1 if the woman was assigned by the lottery as a beneficiary of the program and zero otherwise. The set of controls is the same as in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

Inferencia Causal

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